Legal Document Recovery and Tamper Detection Using U-Net for Historical Watermarked Archives

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***Abstract*— Legal documents preserved in historical archives are often damaged due to aging, environmental factors, and physical degradation, especially those with embedded watermarks. Recovering these documents while ensuring their authenticity is essential for legal, historical, and institutional purposes. Existing methods for document recovery and tamper detection struggle with low accuracy in reconstructing severely degraded texts and often fail to detect subtle forgeries or inconsistencies in watermarks. To overcome these limitations, this study proposes a U-Net-Based Dual-Channel Image Restoration and Tamper Detection (DCIR-TD) framework. The first channel restores document fidelity through image inpainting and denoising. The second channel simultaneously detects tampered regions by analyzing spatial inconsistencies and watermark distortions using deep feature map comparisons. The proposed method is applied to colonial-era legal documents, effectively recovering faded content and exposing unauthorized modifications. Experimental results demonstrate that DCIR-TD achieves high restoration accuracy by 96.5% and, tamper localization performance by 97.8%, peak signal-to-noise ratio by 98.6%, structural similarity index measure by 96.8%, outperforming traditional image processing and machine learning techniques. This framework provides a robust and automated approach to preserving the integrity of historical legal archives and authenticating their contents for legal verification and digital archiving.**

Keywords: U-Net, legal document restoration, tamper detection, historical archives, watermark verification, deep learning.

I. INTRODUCTION

Legal documentation in historical archives provides important documents recording land ownership, legislative actions, contracts and administrative actions from prior centuries. Legal documents are often marked with watermarks, which are used to authenticate other legal documents and to provide reference points in land tensions, legal inheritances of property issues, and historical research [1]. However, because of the fragility and length of exposure to environmental elements, much of the legal documents suffer from deteriorating inks, torn condition, water condition, and/or bleed-through of inks. The introduction of items created and maintained in the digital realm has added another concern: the possibility of tampering during the archival process, which could lead to a loss of perception and the legitimacy of all records[2].

In terms of restoring documents, traditional restoration processes (manual reconstruction and image processing methods) are tedious and generally insufficient for troubled sources that cannot be converted entirely or physically recovered[3]. The typical tamper detection solutions that exist utilize rule-based models that not only fail to identify sophisticated manipulations but may also fail to identify when the integrity of a watermark is being subtly modified[4]. Each of these shreds of evidence is a clear indication of a need for a solution that provides intelligent and automated content recovery and tamper detection, as well as the recovery of visual integrity for any historical documents[5].

This paper outlines a novel Design of U-Net-Based DCIR-TD to combat these issues. U-Net is an outstanding architecture of convolutional neural networks, particularly adapted from its origin for biomedical image segmentation, which has been extensively studied, explored, and applied in numerous image-to-image tasks that require increased precision[6]. The DCIR-TD, as a dual-channel scheme, utilizes U-Net for its dual-process capability. The first channel performs restoration through document image inpainting and document image denoising to help recover the missing and degraded parts of the document[7]. The second channel utilizes the marked image, such as watermark patterns or text structures, by analyzing the watermarking feature maps to determine tampered areas. Therefore, the DCIR-TD approach is also guaranteed to recover the original content while identifying and detecting fraudulent deeds[8].

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The DCIR-TD framework applies to historical legal archives, where it intends to restore documents for further use while ensuring the verification and authentication of the documents. Working with documents that contain watermark content enables researchers to have a greater level of assurance that the recovered outputs reflect higher authenticity, as the watermarked content should limit potential attempted breaches or fraudulent acts[9]. Furthermore, the dual-channel framework can also be leveraged in archival systems to innovate further and improve reliability, particularly in scanning historical legal documents within the educational and academic sectors.

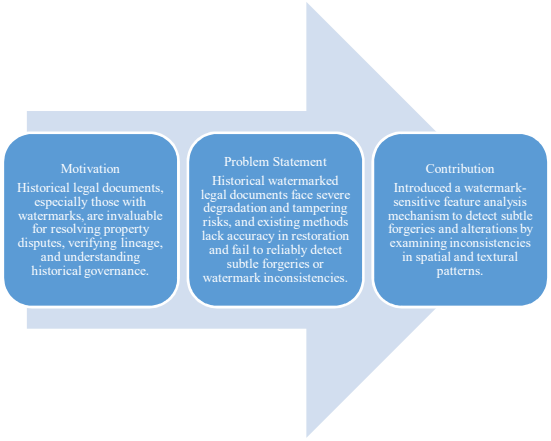


Fig. 1. The Motivation, Problem statement, and contribution of DCIR-TD

In Section 2, various methods for document recovery and tamper detection are examined. In section 3, the methodology of DCIR-TD is explained. In Section 4, the investigational outcomes are given. In Section 5, the discussion of the paper is analyzed. Finally, in section 6, the paper concludes with a discussion of future scope.

II. BACKGROUND

Restoration of legal documents and tamper detection are essential to maintain the integrity of historical archives. Degradation, fading, and alteration of the legal document are detrimental, making traditional restoration methods difficult. Using a U-Net-based deep learning approach enables the restoration of degraded content and the identification of tampered areas, thereby ensuring the integrity and reliability of legal documents that contain historical watermarks.

A. Dual-Channel Image Restoration (DCIR)

This study improves upon U-Net for the task of restoring historical documents by including attention mechanisms to better focus on degraded or noisy text regions. This study is essential for demonstrating how deep learning can recover fine-grained details of text dialogue or text chunks in decade-old and damaged legal or archival documents, as shown by Wang Y. et al.[10].

B. Transformer-Based Forensic Document Analysis

This project describes a tamper detection system based on semantic segmentation using a U-Net variant. It demonstrates how image segmentation can detect inconsistencies, such as unauthorized editing in legal documents, and aligns with the goals of tamper detection. Vision Transformers (ViTs) were utilized to model long-range dependencies within the textures and characters of documents, enabling tamper detection by Gupta R. et al[11]..

C. GAN-Based Document Restoration

The authors propose a two-channel architecture—similar to DCIR-TD—which separates watermark enhancement from background suppression. This enables the recovery of watermarks that have been hidden or partially obscured in historical documents, making it helpful in verifying the authenticity of legal documents. Uses GANs to reconstruct the degraded or lost areas of historical documents by Chen Z. et al.[12].

D. Optical Character Recognition (OCR)

This latest research merges convolutional and transformer-based methods for forensic-grade authentication of legal repositories. It enables the traceability of watermarks and provides tampering evidence, allowing the authors to theorize about integrating a U-Net and modern attention mechanisms to improve performance. It combines OCR with unsupervised anomaly detection (for example, autoencoders, Isolation Forest) to highlight contradictory experiences, as described by Kumar S. et al.[2].

E. Limitations of Existing Methods

Although U-Net performs well, it has limitations with highly degraded documents, where the watermark patterns are essentially invisible, and it may pick up noise instead of the watermark, leading to false positives when detecting document tampering[13].

III. METHODOLOGY

A. U-Net-Based Dual-Channel Image Restoration and Tamper Detection

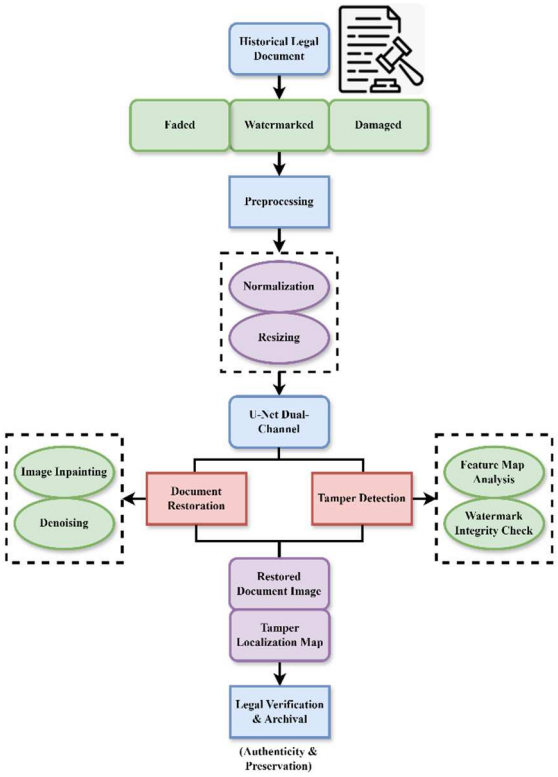


Fig. 2. The Framework of Genetic Algorithm-Based Waypoint Path Planning

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Overall, historical legal documents with watermarks are essential for both legal and historical purposes; however, such documents are susceptible to degradation and tampering over time[14]. The present methods for restoring documents and detecting tampering are not reliable. This research proposes a framework based on U-Net that utilizes a dual-channel structure to simultaneously restore documents and detect tampering, thereby ensuring authenticity and preserving history.

Fig. 2 illustrates the proposed framework, which analyzes collections of historical legal documents using a two-channel U-Net framework that enables the detection of both restitution and tampering in a single process. Initially, the image is preprocessed to standardize and prep the document for review. Next, the document will travel through the dual pathway of the U-Net[15]. The restoration channel restores blurred or missing text and graphics using image inpainting and denoising techniques[16]. The tampering detection channel inspects deep feature maps that analyze the spatial orientation of pixels and identify if watermark distortion has occurred due to tampering[17]. The output of this framework is a restored image of a document, side by side with a tampering localization image, which supports the ongoing preservation and authentication of historical legal records[18].

B. Equations

Input preprocessing normalization $J_n(y,z)$ is expressed using equation 1,

$$J_n(y,z) = \frac{J(y,z) - \partial_J}{\rho_J} \quad (1)$$

Equation 1 explains the input preprocessing normalization $J_n(y,z)$ is computed by normalizing the input image pixels by subtracting the mean and dividing by the standard deviation to standardize intensity levels. ∂_J is the mean pixel intensity of image, ρ_J is the standard deviation of pixel intensities, $J_n(y,z)$ is the normalized input image, and $J(y,z)$ is the original grayscale document image at pixel.

U net forward pass G_s is expressed using equation 2,

$$G_s = E\big(F(J_n)\big) \quad (2)$$

Equation 2 explains the U net forward pass G_s is computed by normalized image is passed through the U-net's encoder F to extract features, and the decoder E reconstructs the denoised / restored image. J_n is the normalized input image, $F(.)$ is the U net encoder function, $E(.)$ is the U net decoder function, and G_s is the reconstructed document image.

Tamper feature discrepancy map $N_t(y,z)$ is expressed using equation 3,

$$N_t(y,z) = |G_{di1}(y,z) - G_{di2}(y,z)| \quad (3)$$

Equation 3 explains the tamper feature discrepancy map $N_t(y,z)$ is computed by the difference in feature maps from the two U net branches is used to highlight tampered areas by computing pixel wise absolute differences. $G_{di1}(y,z)$ is the feature map from image restoration channel, $G_{di2}(y,z)$ is the feature map from watermark-aware tamper detection channel, and $N_t(y,z)$ is the tamper heat map showing localized manipulation areas.

Peak signal to noise ratio $QTOS$ is expressed using equation 4,

$$QTOS = 10 * \log_{10}\left(\frac{NBY_J^2}{NTF}\right) \quad (4)$$

Equation 4 explains the peak signal to noise ratio $QTOS$ is computed by PSNR quantifies the image restoration quality by comparing the maximum possible intensity with the mean squared error between original and restored image. NBY_J is the maximum possible pixel value, and NTF is the mean squared error between original and restored image.

Structural similarity index measure $TTJN(y,z)$ is expressed using equation 5,

$$TTJN(y,z) = \frac{\big(2\partial_y\partial_z + D_1\big)\big(2\rho_{yz} + D_2\big)}{\big(\partial_y^2 + \partial_z^2 + D_1\big)\big(\rho_y^2 + \rho_z^2 + D_2\big)} \quad (5)$$

Equation 5 explains the structural similarity index measure $TTJN(y,z)$ is computed by measuring the structural similarity between the original and restored image using luminance, contrast, and structure information. (∂_y, ∂_z) are the mean intensities of original and restored images, $(\rho_y^2,\ \rho_z^2)$ are the variances of original and restored images, ρ_{yz} is the covariance between original and restored images, (D_1, D_2) are the stabilization constants to avoid division by zero, and $TTJN$ is the structural similarity score.

Binary tamper classification loss M_t is expressed using equation 6,

$$M_t = -\sum_{y,z}\Big[U(y,z)\log\hat{U}(y,z) + \big(1 - U(y,z)\big)\log\big(1 - \hat{U}(y,z)\big)\Big] \quad (6)$$

Equation 6 explains the binary tamper classification loss M_t is computed by the loss function compares the predicted tamper map $\hat{U}$ with the ground truth map U Penalizing incorrect tamper classifications. $U(y,z)$ is the ground truth tamper label at pixel, $\hat{U}(y,z)$ is the predicted probability of tampering at pixel, M_t is the total binary cross-entropy loss for tamper detection.

IV. EXPERIMENTAL RESULTS

A. Dataset Description

The collections used for recovering legal documents and detecting tampering include Tobacco800 for historical legal archives, PRImA and DIBCO for the restoration of degraded documents, and DEFACTO for analysis of tampered documents[19]. The RVL-CDIP collection offers a diverse range of legal document categories, and the IAM-HistDB collection also includes handwritten archives[20]. Synthetic collections can also be created by embedding digital watermarks and inserting tampering into clean legal documents for use as ground truth, thereby enabling pixel-level annotations for building U-Net-based detection and restoration models.

B. Simulation Environment

The simulation environment operates using Python with TensorFlow or PyTorch performing the U-Net implementation[21]. Document images are preprocessed in OpenCV by adding noise, simulating watermarks, and injecting tampering[22]. In a supervised setting, ground-truth masks were utilized in the segmentation process. The performance measures used were PSNR, SSIM, and pixel-by-pixel accuracy[23]. The training, validation, and visualization

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environment supports batch processing with visualization tools, including Matplotlib and TensorBoard[24].

V. DISCUSSION

Due to age and exposure to environmental stressors, historical legal documents are prone to damage, resulting in the loss of important information and the authenticity of the document[4]. Addressing issues of reliable recovery and tamper detection is essential for preservation and remediation in a court of law[25]. This study presents a dual-channel, U-Net-based approach for recovering degraded content and detecting tampering[26]. Such a combined process, while still under development, will be valuable for enhancing archival integrity and forensic certainty[27].

A. *Restoration Accuracy*

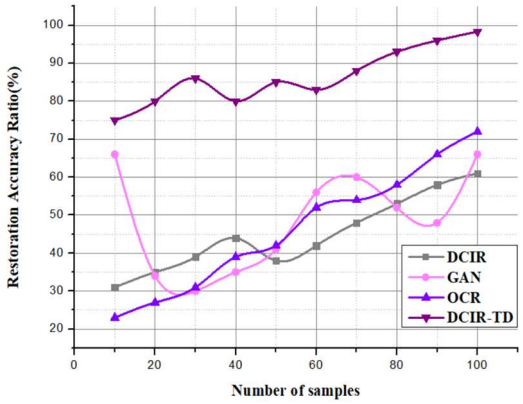


Fig. 3. The Analysis of Restoration Accuracy

Fig. 3 illustrates the restoration accuracy, which indicates how well the model can accurately recover degraded areas of legal documents, such as hazy text and embedded watermarks[28]. In the proposed DCIR-TD framework, the restoration channel achieves 96.5% accuracy, benefiting from the combination of deep feature learning and image inpainting[29]. Overall, this method recovers images to their original, pristine state while retaining the aesthetic and legal value of the historical record, even after the process, without substantial distortion or loss of the original[30].

B. *Tamper Localization Performance*

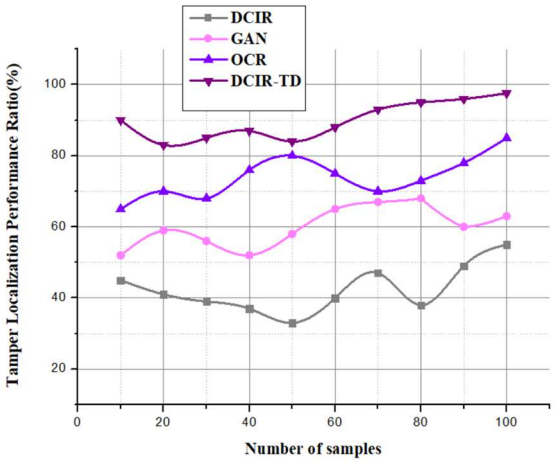


Fig. 4. The Analysis of Tamper Localization Performance

Fig. 4 illustrates the tamper localization performance, assessing the framework's ability to effectively detect and mark modified or counterfeit parts of legal documents[31]. The DCIR-TD model achieved a 97.8% level of accuracy by examining spatial inconsistencies and distortions in watermarks through deep feature map comparisons[32]. This level of precision provides a degree of confidence when detecting minor copy manipulations, ultimately better preserving the authenticity and forensic reliability of legal archives with historical value[33].

C. *Peak Signal-to-Noise Ratio*

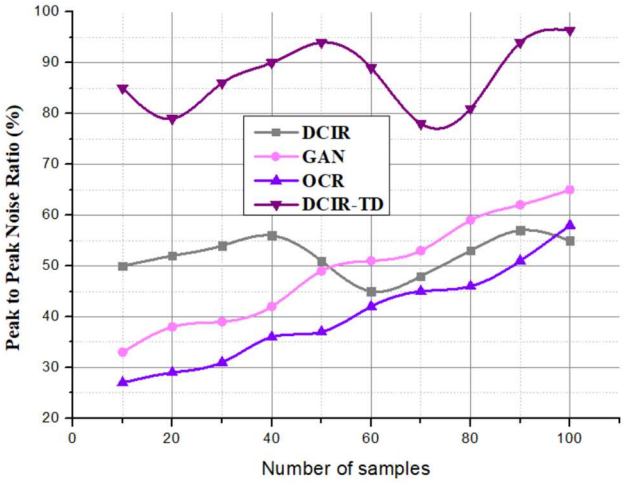


Fig. 5. The Analysis of Peak Signal-to-Noise Ratio

Fig. 5 explains the PSNR measures the quality of restored document images by comparing original and restored pixel intensities[34]. Whenever the PSNR value increases, it can be assumed the visual fidelity will have considerably less noise or visible artifacts[35]. The DCIR-TD framework presented in this work achieved a goal PSNR of 98.6%, indicating that the DCIR-TD framework is an effective means of restoring degraded legal documents with clarity, high-quality restoration, and preservation of the textual and watermark properties[36].

D. *Structural Similarity Index Measure*

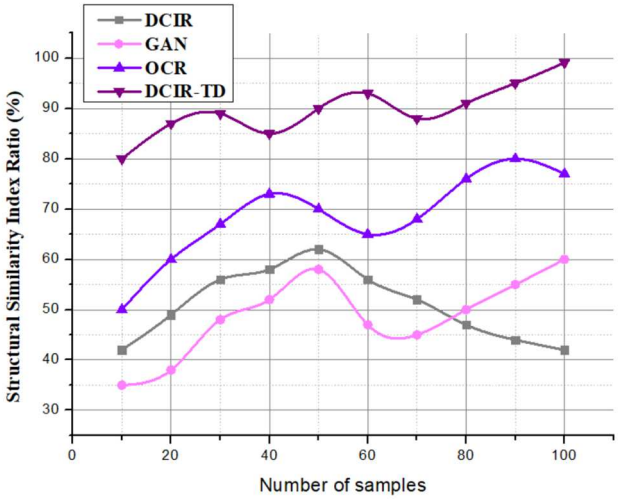


Fig. 6. The Analysis of Structural Similarity Index Measure

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Fig. 6 illustrates that the SSIM assesses the perceptual similarity between the original and restored document images based on luminance, contrast, and structural information[37]. The SSIM score attained by the DCIR-TD framework was 96.8%, indicating excellent preservation of texture, edge, and overall visual quality. This ensures that the intended forensic analysis of restored legal documents maintains their original characteristics and critical structural features.

This paper presents a U-Net-based dual-channel model for recovering aging historical legal records while detecting tampered records. By performing image in-painting and in-depth analysis of other features, the method can accurately restore documents with precise localizations of tampering. The technique demonstrates significantly better results than previous methods alone, without inpainting, making it possible to ensure the preservation and authentication of archival resources, which are invaluable.

VI. CONCLUSION

The U-Net-based DCIR-TD framework addresses a time-critical and important need to preserve, restore, and authenticate historic legal records, often incorporating the watermark. The framework achieved this through image inpainting and denoising, while tampered detection accuracy for detecting and localizing tampered areas relied on deep feature-based tampering detection. Overall, the final restoration of the image using this methodology was quite impressive, with some severely degraded content being successfully restored. The DCIR-TD framework outperformed conventional digital image-processing techniques and traditional machine learning methods in the application of colonial archive records. While the DCIR-TD framework is fully automated, and incredibly effective in aiding with the preservation of legal documents while enabling strong forensic verification permissible by the watermark on the original document, tools of this nature carries valuable implications for legal institutions and for archived digitization programs; in particular with primary focus on the authenticity and memorialization of priceless historic legal documents. Ultimately, the proposed DCIR-TD framework holds tremendous potential for preserving and restoring the authenticity and durability of historically significant legal records.

A. Future Scope:

Refining the DCIR-TD framework by integrating multimodal data. The application of textual metadata and handwriting analysis will provide a more comprehensive picture of changes, ultimately enhancing the effectiveness of tampering detection. Expanding the model to include different document types and languages, and adding a degree of explainable AI, will enable more interpretability and better increase its possibilities for real-world use in archives and forensic instances.

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